DEPT. OF ELECTRICAL ENGG. – SESSIONAL EXAM-I B.TECH. ELECT. ENGG. 3RD SEM

LINEAR SYSTEM THEORY: – EEPC206: E1-E3

Note: Attempt all questions

Max Marks: 15 Time: 50 minutes

1. Find the orthogonal basis for W, where $W = span\{x_1, x_2\}$ $x_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ and $x_2 = \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix}$. (2)
2. State two properties of Symmetric matrices. (1)
3. Prove that any matrix A can be written as $A = U\Sigma V^T$, where U and V are the left and right singular vectors of $A^T A$. (3)
4. Given $G(s) = \frac{s+4}{s^2+7s+21}$. Realize $G(s)$ into diagonal form. (3)
5. Using Cayley Hamilton Theorem compute the state transition matrix $\phi(t)$ for diagonal form obtained in Q.4 (3)
6. Show that there exist infinitely many state-space representation for a given system $G(s)$ (3)